

Part 3: Multivariate Distributions

Exercise: Suppose that X and Y have pmf's

$$f_X(x) = \begin{cases} 0.5, & x = 100 \\ 0.5, & x = 250 \end{cases}$$

and

$$f_Y(y) = \begin{cases} 0.25, & y = 0 \\ 0.25, & y = 100 \\ 0.5, & y = 200 \end{cases}$$

What is $P(X > Y)$?

[illegible]

In the case of a pair of discrete random variables X and Y , we need to specify the **joint probability mass function** in order questions such as the previous one. We naturally write

$$f_{X,Y}(x,y) = P(X = x, Y = y)$$

Note that $f_{X,Y}(x,y) \geq 0$ for all x and y and that

$$\sum_x \sum_y f_{X,Y}(x,y) = 1$$

where the sum is over all pairs (x, y) for which $f_{X,Y}(x, y) > 0$.

The CDF is defined as

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

The extension to more than two random variables is simple.

Exercise: A large insurance agency services a number of customers who have purchased both a homeowner's policy and an automobile policy from the agency. For each type of policy, a deductible amount must be specified. Let X be the deductible amount on the auto policy, and Y the deductible amount on the homeowner's policy. The joint pmf of (X, Y) is given by the following table:

x	y		
	0	100	200
100	.20	.10	.20
250	.05	.15	.30

What is the probability that X is larger than Y ?

The Continuous Case

The pair of random variables (X, Y) is **jointly continuous** with density $f_{X,Y}(x, y)$ if

1. $f_{X,Y}(x, y) \geq 0$ for all x, y

2.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$$

To calculate probabilities, integrate the pdf over the region of interest:

$$P((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy$$

Here, it is **not** the case that A need be rectangular.

A useful property: In the continuous case,

$$\frac{\partial F_{X,Y}(x, y)}{\partial x \partial y} = f_{X,Y}(x, y)$$

1 **Exercise:** Suppose that the joint pdf of (X, Y) is given by:

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$$f_{X,Y}(x, y) = \begin{cases} 24xy, & x > 0, y > 0 \\ & x + y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

- 3 1. What is the probability that $X + Y$ is at most 0.5?
- 4 2. Find $P(X > Y)$.
- 5 3. Find the pdf of the random variable $T = X + Y$.

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1 **The Beta(α, β) Distribution**

2 In the previous Exercise, the random variable T has the Beta($\alpha = 4, \beta = 1$)
3 Distribution.

4 The random variable X has the **Beta(α, β) Distribution** if it has density

$$5 \quad f_X(x) = \begin{cases} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

6 This is a useful model for proportions or probabilities.

7 Note that $E(X) = \alpha/(\alpha + \beta)$.

8 Also note that if $\alpha = \beta = 1$, then the distribution is Uniform(0,1). Further,
9 if $\alpha = \beta$ then the distribution is symmetric about 0.5.

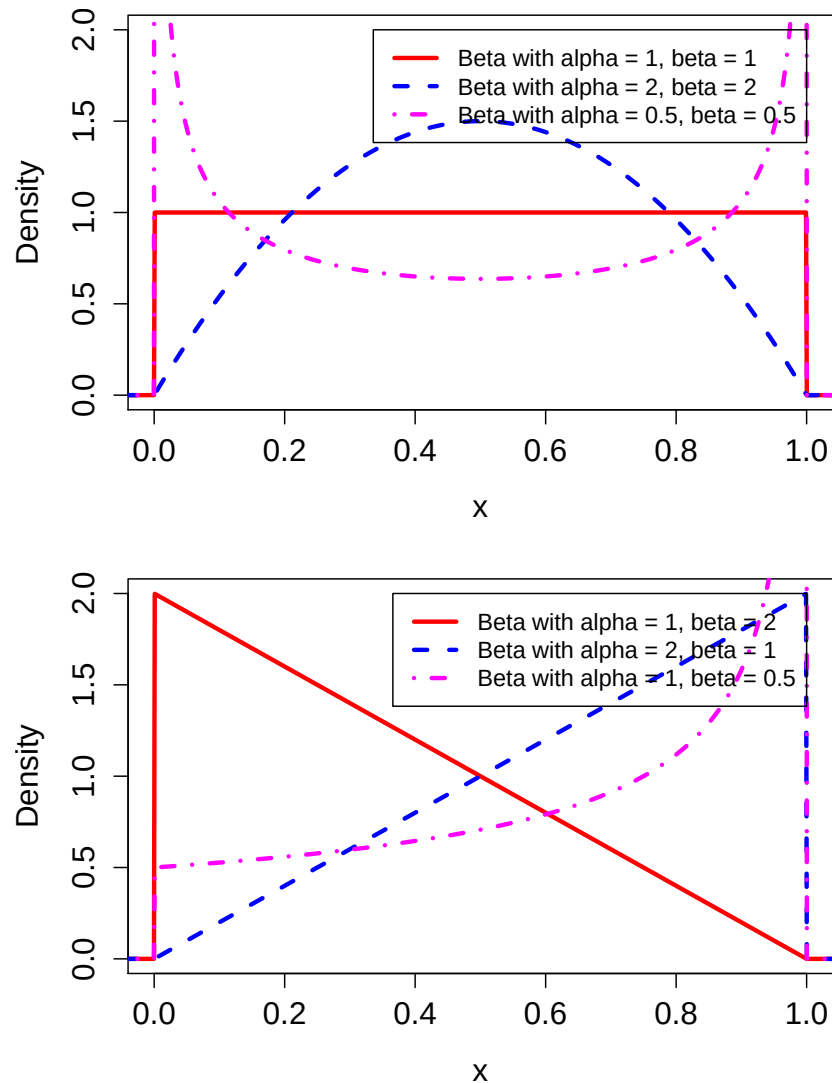


Figure 1: Beta Distribution pdfs.

Marginal Distributions

If one starts with the joint distribution (pmf or pdf) for (X, Y) , it is possible to derive the distribution for X ignoring Y and the distribution for Y ignoring X .

These are called the **marginal distributions** for X and Y , respectively.

These are simply the f_X and f_Y already described.

In the **discrete case**, you “sum over” the other random variable, i.e.,

$$f_X(x) = P(X = x) = \sum_y P(X = x, Y = y) = \sum_y f_{X,Y}(x, y)$$

Exercise: Find f_X and f_Y in the Insurance example above.

In the **continuous case**, you “integrate out” the other variable:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

Exercise: Return to our “triangular support” example. Find f_X and f_Y .

Exercise: An electronic system has one each of two different types of components in joint operation. Let X and Y denote the random lengths of life of the components of type I and II, respectively. The joint function is:

$$f_{X,Y}(x,y) = \begin{cases} 8xe^{-2(x+y)} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the marginal pdf $f_X(x)$ and $f_Y(y)$.

In the above example, we can see that

$$f_{X,Y}(x, y) = f_X(x)f_Y(y), \text{ for all } x, y$$

This implies that X and Y are **independent**.

The above can be taken as the definition of independent random variables in both the discrete and continuous cases. Further, we say that $X_1, X_2, \dots, X_n$ are independent if for all $x_1, x_2, \dots, x_n$,

$$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = f_{X_1}(x_1)f_{X_2}(x_2) \cdots f_{X_n}(x_n)$$

Exercise: Consider a case where (X, Y) have a continuous bivariate distribution. Is “the support of the distribution is rectangular” a necessary or a sufficient condition for X and Y being independent?

Rule for the Lazy Statistician

For a general function $g(x, y)$, one can find expected values using

$$E(g(X, Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y)f_{X,Y}(x, y) dx dy$$

Exercise: Try the case $g(x, y) = x$ to verify that the result is $E(X)$.

Exercise: Assume that X and Y are independent.

Show that $E(h_1(X)h_2(Y)) = E(h_1(X))E(h_2(Y))$.

Exercise: Assume that U_1 and U_2 are independent uniform(0, 1) random variables. Let $U_{(1)}$ denote the minimum of U_1 and U_2 , and let $U_{(2)}$ denote the larger of those two. Finally, let

$$L_1 = U_{(1)}, \quad L_2 = U_{(2)} - U_{(1)}, \quad \text{and} \quad L_3 = 1 - U_{(2)}.$$

Note that this is the “stick-broken-into-three-pieces” example, with L_i denoting the lengths of the three pieces ordered from left to right.

1. What is the joint distribution of (U_1, U_2) ?
2. What is the joint distribution of $(U_{(1)}, U_{(2)})$?
3. What is the joint distribution of (L_1, L_2) ?
4. Show that L_1 , L_2 , and L_3 each have the Beta(1, 2) distribution. (Do not take it as “obvious” that all three have the same distribution.)

Comment: It is not difficult to see that if k uniform cuts are made, the length of each piece has the Beta(1, k) distribution.

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The Conditional Distributions

If we know (condition on) $Y = y$, what is the distribution of X ? Such a question is answered by considering the **conditional distribution** of X given Y .

In the **discrete case**,

$$f_{X|Y}(x|y) = P(X = x|Y = y) = \begin{cases} f_{X,Y}(x, y) / f_Y(y), & f_Y(y) > 0 \\ 0, & f_Y(y) = 0 \end{cases}$$

and in the **continuous case** we have

$$f_{X|Y}(x|y) = \begin{cases} f_{X,Y}(x, y) / f_Y(y), & f_Y(y) > 0 \\ 0, & f_Y(y) = 0 \end{cases},$$

with $f_{X|Y}$ called the **conditional density of X given Y** .

Comment: The continuous case is notable in that the event being conditioned on ($Y = y$) has probability zero. There is nothing mathematically or conceptually incorrect about conditioning on probability zero events.

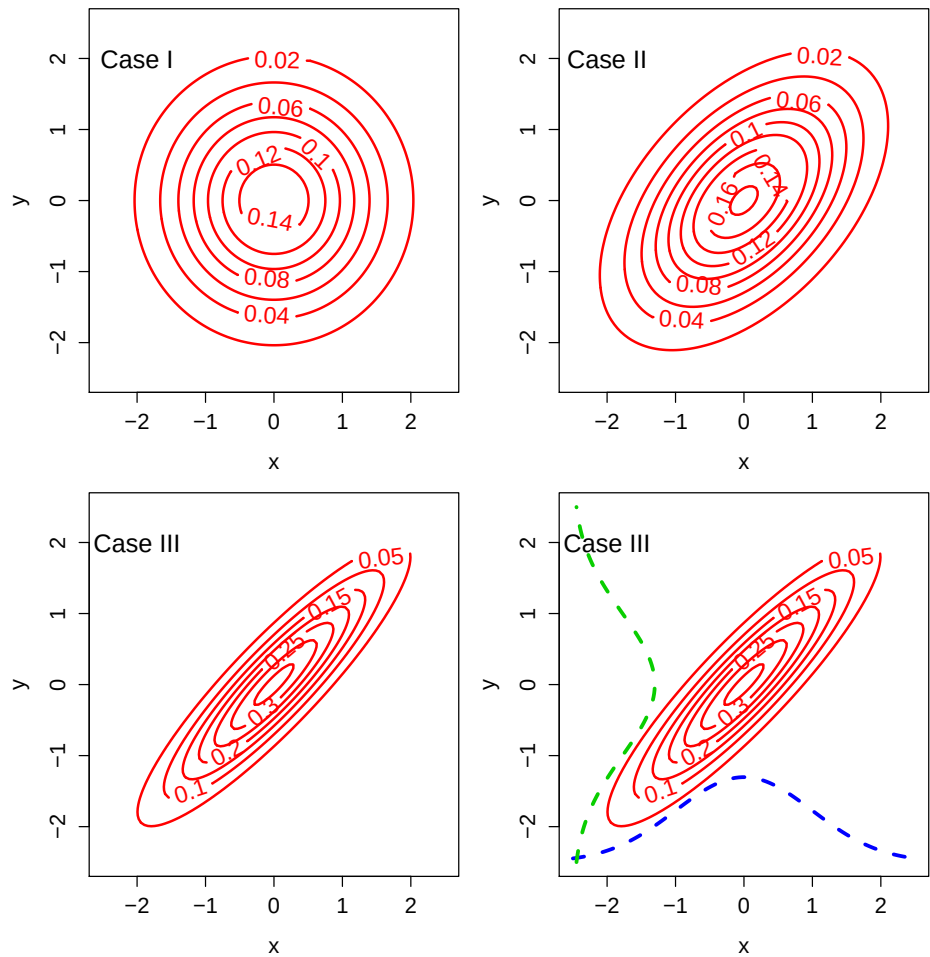
1 **Exercise:** Derive the conditional distribution of X given Y in our “triangular support” example.
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1 **Example:** This example illustrates the distinction between the marginal and
2 conditional distributions. Here, the bivariate pdf is depicted using **contour**
3 **plots**.

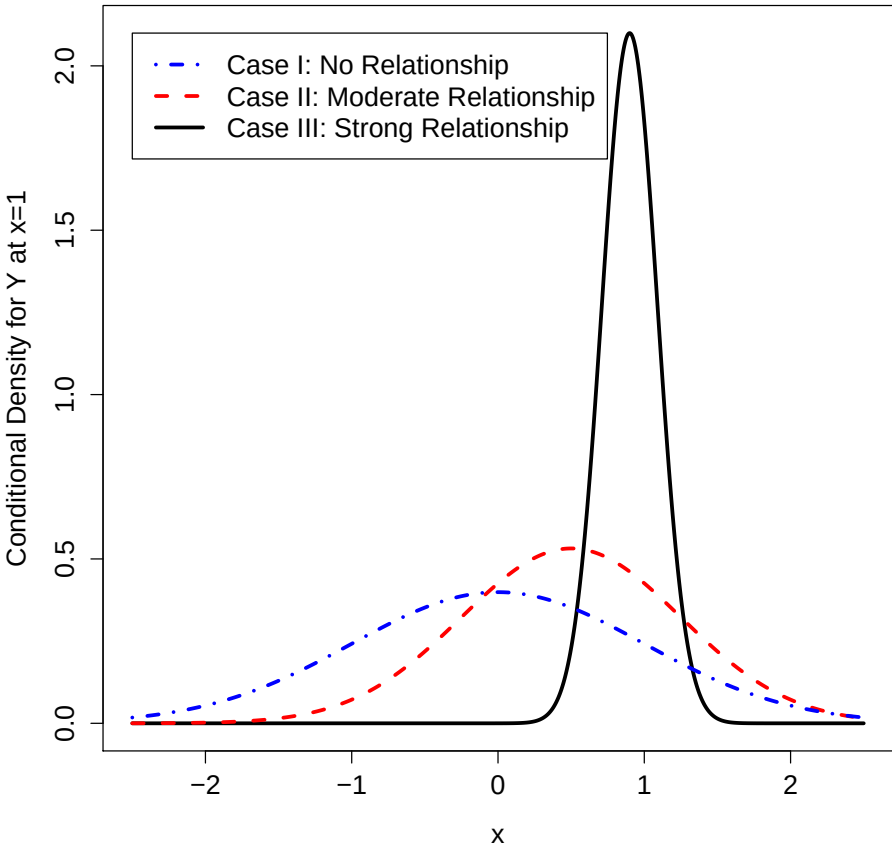
4 In **Case I** shown below, X and Y are independent, and every conditional
5 distribution is the same. In **Case II** there is a moderate amount of depen-
6 dence. In **Case III** the dependence is stronger.

7 The fourth plot again shows Case III, but with the marginals imposed. In
8 fact, the marginals are the same in all three Cases.



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The conditional distribution Y for $X = 1$ changes across the three Cases.



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Covariance

The **covariance between X and Y** , denoted $\text{Cov}(X, Y)$, is defined as

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

This is a measure of the amount of **linear association** between X and Y .

The sign of $\text{Cov}(X, Y)$ tells us the **direction** of that association.

Exercise: Looking at the definition of the covariance, explain under what conditions $\text{Cov}(X, Y) > 0$, $\text{Cov}(X, Y) < 0$, and $\text{Cov}(X, Y) = 0$.

1 When $\text{Cov}(X, Y) = 0$, then X and Y are said to be **uncorrelated**.

2 Further, the **correlation between X and Y** is defined to be

$$3 \quad \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{V(X)V(Y)}}$$

4 Properties that you should know:

5 1. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$

6 2. $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$

7 3. $\text{Cov}(X, X) = V(X)$

8 4. $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$

9 5. If X and Y are independent, then $\text{Cov}(X, Y) = 0$.

10 6. $-1 \leq \text{Corr}(X, Y) \leq 1$

1 **Exercise:** Find $\text{Cov}(X, Y)$ for the “triangular support” example.

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Exercise: Suppose that X and Y are random variables with $Y = aX + b$ for constants a and b with $a \neq 0$. What is the correlation between X and Y ?

What is a Matrix?

A *matrix* is an array of numbers in a rectangular form.

Examples:

$$\mathbf{A} = \begin{bmatrix} 3 & 5 \\ 1 & 4 \\ 2 & 6 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{bmatrix}$$

The *dimension* of a matrix is the number of rows by the number of columns. The dimension of $\mathbf{A}$ is 3 by 2. The dimension of $\mathbf{B}$ is 3 by 4.

In math notation, a matrix is written:

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \\ x_{41} & x_{42} & x_{43} \end{bmatrix}$$

or

$$\mathbf{X} = [x_{ij}]$$

where i is the row number, j the column number.

In the above examples, $a_{12} = 5$, $a_{31} = 2$, $b_{32} = 4$, $b_{24} = 5$.

The individual numbers are *elements* of the matrix.

Special Matrices

A *vector* is a matrix that has n rows and 1 column.

$$\mathbf{u} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

(A row vector has 1 row and n columns.)

$$\mathbf{v} = \begin{bmatrix} 3 & 1 & 2 \end{bmatrix}$$

A *square* matrix has the same number of rows and columns.

$$\mathbf{S} = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}$$

A *symmetric* matrix has elements such that $a_{ij} = a_{ji}$.

$$\mathbf{S} = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 4 & 6 \\ 2 & 6 & 5 \end{bmatrix}$$

So a symmetric matrix must also be a square matrix.

A *diagonal* matrix is a matrix that is zero everywhere except on the diagonal - the elements whose row number equals their column number $\{(1,1), (2,2), (3,3), \text{etc}\}$.

$$\mathbf{D} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

A special case of a diagonal matrix is the *identity matrix*.

Its diagonal elements are all ones.

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The identity matrix (or any other diagonal matrix) is also a symmetric matrix.

Matrix Operations

Matrix Equality: Two matrices $\mathbf{A}, \mathbf{B}$ are equal if and only if, for all elements, each $a_{ij} = b_{ij}$. (Note: so must have same dimensions.)

Matrix Transpose: The *transpose* of a matrix is found by interchanging the corresponding rows and columns of a matrix. The first row becomes the first column, the second row becomes the second column, etc. The dimensions are then switched and the element a_{ij} becomes the element a_{ji} . The transposed matrix is often denoted $\mathbf{A}'$ (or $\mathbf{A}^T$).

$$\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 1 & 4 \\ 2 & 6 \end{bmatrix}$$
$$\mathbf{A}' = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 4 & 6 \end{bmatrix}$$

Matrix Addition or Subtraction:

Two matrices can be added or subtracted only if their dimensions are the same (both rows and columns). The corresponding elements are then added or subtracted.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$
$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \end{bmatrix}$$

Matrix Multiplication:

To multiply a matrix by a *scalar* (just a constant number), just multiply each element by that number.

$$\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 1 & 4 \\ 2 & 6 \end{bmatrix} \quad 4\mathbf{A} = \begin{bmatrix} 12 & 4 \\ 4 & 16 \\ 8 & 24 \end{bmatrix}$$

Two matrices can be multiplied only if the number of columns of the first matrix equals the number of rows of the second matrix (the inside numbers). The resulting matrix has dimension: number of rows of first matrix by number of columns of second matrix (the outside numbers).

Examples:

$\mathbf{A}$ is (3 by 2); $\mathbf{B}$ is (2 by 4).

$\mathbf{AB}$ (3 by 2 · 2 by 4) is allowed; dimension (3 by 4).

$\mathbf{BA}$ (2 by 4 · 3 by 2) is not allowed.

We find each element $(ab)_{ij}$ by summing the crossproducts of the i th row of $\mathbf{A}$ and the j th column of $\mathbf{B}$.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$$
$$\mathbf{AB} = \begin{bmatrix} a_{11} \cdot b_{11} + a_{12} \cdot b_{21} + a_{13} \cdot b_{31} & a_{11} \cdot b_{12} + a_{12} \cdot b_{22} + a_{13} \cdot b_{32} \\ a_{21} \cdot b_{11} + a_{22} \cdot b_{21} + a_{23} \cdot b_{31} & a_{21} \cdot b_{12} + a_{22} \cdot b_{22} + a_{23} \cdot b_{32} \end{bmatrix}$$

Example:

$$\mathbf{A} = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 3 & 5 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 2 & 0 \\ 4 & 1 \\ 6 & 2 \end{bmatrix}$$

IMPORTANT: $\mathbf{AB}$ is not necessarily equal to $\mathbf{BA}$.
Multiplication order matters!!

Note: any (square) matrix $\mathbf{A}$ multiplied by the identity matrix $\mathbf{I}$ is just the original matrix $\mathbf{A}$.

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{A} = \begin{bmatrix} 2 & 0 & 5 \\ 4 & 1 & 2 \\ 6 & 2 & 3 \end{bmatrix}$$

$$\begin{aligned} \mathbf{IA} &= \begin{bmatrix} 1 \cdot 2 + 0 \cdot 4 + 0 \cdot 6 & 1 \cdot 0 + 0 \cdot 1 + 0 \cdot 2 & 1 \cdot 5 + 0 \cdot 2 + 0 \cdot 3 \\ 0 \cdot 2 + 1 \cdot 4 + 0 \cdot 6 & 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 2 & 0 \cdot 5 + 1 \cdot 2 + 0 \cdot 3 \\ 0 \cdot 2 + 0 \cdot 4 + 1 \cdot 6 & 0 \cdot 0 + 0 \cdot 1 + 1 \cdot 2 & 0 \cdot 5 + 0 \cdot 2 + 1 \cdot 3 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 0 & 5 \\ 4 & 1 & 2 \\ 6 & 2 & 3 \end{bmatrix} = \mathbf{A} (= \mathbf{AI}) \end{aligned}$$

Exercise: In Probability and Statistics, one will frequently encounter products of vectors and matrices of the form $\mathbf{u}^T \mathbf{A} \mathbf{u}$, where $\mathbf{u}$ is a column vector with n rows and $\mathbf{A}$ is a n by n matrix. Write out this product in a way that avoids the use of matrices.

Linear Dependence:

You have a matrix $\mathbf{A}$ with 2 columns. For ex:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Examine the columns. The second column is twice the first column: $1 \cdot 2 = 2$; $2 \cdot 2 = 4$. The second column does not add to the information in the matrix; think of it like a copy.

If a column can be written as a linear combination of any other columns in the matrix, the columns are linearly dependent.

Are these linearly dependent?

$$\mathbf{B} = \begin{bmatrix} 1 & 4 & -3 \\ 6 & 22 & -18 \\ 3 & -2 & -9 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 2 & 8 \\ 6 & 4 & 8 & 12 \\ 3 & 5 & 1 & -20 \end{bmatrix}$$

Rank of Matrix: maximum number of linearly independent columns in a matrix. If all columns are linearly independent, the matrix has *full rank*.

Matrix Inverse: The inverse of a matrix $\mathbf{A}$ is the matrix $\mathbf{A}^{-1}$ that satisfies $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$. It also holds that $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$. So you can think of the inverse as the reciprocal of a matrix. (Only square matrices can have inverses, but not all do...)

Remember that matrix multiplication is not just multiplying pairs of elements, so we can't just find the reciprocal of each element. So how do we find the inverse? How do we even know the inverse exists?

Determinant of a Matrix:

The determinant is a measure, in a sense, of the “volume” of the matrix.

For a 2 by 2 matrix,

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

the determinant is $|\mathbf{A}| = a \cdot d - b \cdot c$.

If it equals zero, there is no “volume” to the matrix; no inverse exists. Matrix is not of full rank.

Example:

$$\mathbf{A} = \begin{bmatrix} 4 & 12 \\ 3 & 6 \end{bmatrix}$$

$|\mathbf{A}| =$

$$\mathbf{A} = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

$|\mathbf{A}| =$

Once we know the inverse exists, we can find it.

For a 2 by 2 matrix,

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \cdot \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

where $|\mathbf{A}| = a \cdot d - b \cdot c$.

Examples:

$$\mathbf{A} = \begin{bmatrix} 2 & 3 \\ 1 & 6 \end{bmatrix}, \quad \mathbf{A}^{-1} = \frac{1}{2 \cdot 6 - 3 \cdot 1} \begin{bmatrix} 6 & -3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} \frac{6}{9} & -\frac{3}{9} \\ -\frac{1}{9} & \frac{2}{9} \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} 4 & 7 \\ 5 & 1 \end{bmatrix}, \quad \mathbf{A}^{-1} = \frac{1}{4 \cdot 1 - 7 \cdot 5} \begin{bmatrix} 1 & -7 \\ -5 & 4 \end{bmatrix} = \begin{bmatrix} -\frac{1}{31} & \frac{7}{31} \\ \frac{5}{31} & -\frac{4}{31} \end{bmatrix}$$

Finding the inverse for higher dimensions involves more complicated formulas and is usually solved by a math software program. (In R, use `solve()`.)

The Bivariate Normal Distribution

The pair (X_1, X_2) has the **bivariate normal distribution** if it has joint pdf

$$f(\mathbf{x}) = \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

where $\boldsymbol{\mu}$ is the vector of means and Σ is the **covariance matrix**:

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_1, X_2) & \text{Var}(X_2) \end{bmatrix}.$$

Some important properties:

1. X_1 and X_2 are normally distributed, i.e. the marginals are normal.
2. The conditional distribution of X_1 given $X_2 = x_2$ is normal with mean

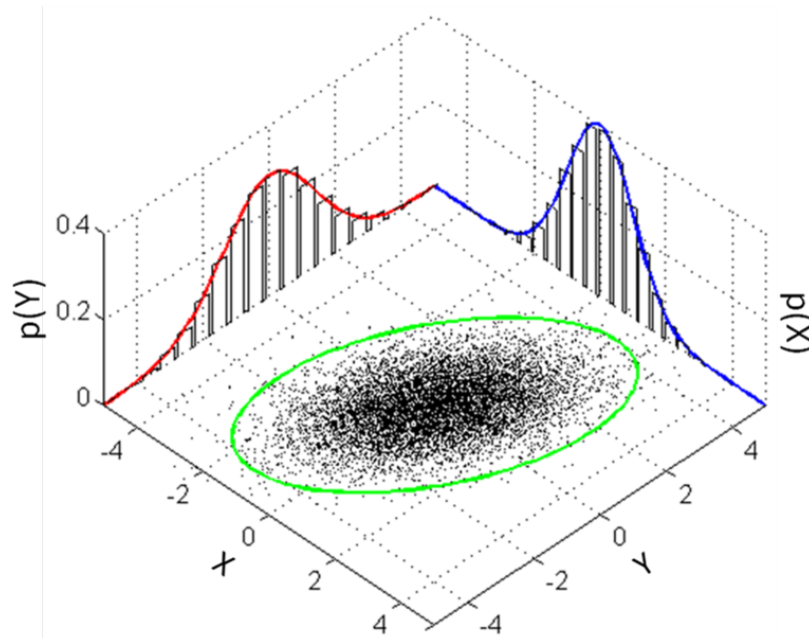
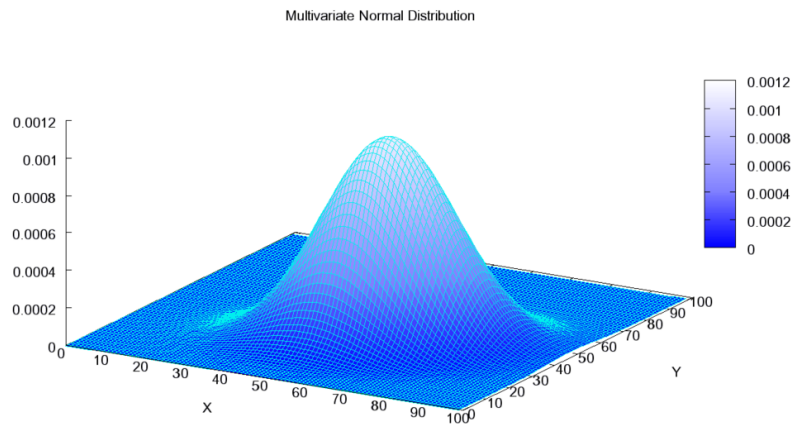
$$\mu_1 + \frac{\sigma_1}{\sigma_2} \rho (x_2 - \mu_2)$$

and variance

$$(1 - \rho^2) \sigma_1^2$$

where ρ equals the correlation between X_1 and X_2 .

3. All linear combinations of X_1 and X_2 are also normally distributed.
4. X_1 and X_2 uncorrelated implies that X_1 and X_2 are independent.



Figures from Wikipedia.

More than Two Random Variables

As one moves to the joint distribution of more than two random variables, it becomes increasingly important to work in matrix notation.

Define a **random vector** of length n by writing

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}$$

where each of the X_i is a random variable. Then we can define

$$E(\mathbf{X}) = \begin{bmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_n) \end{bmatrix}$$

and $V(\mathbf{X})$ is a n by n matrix whose (i, j) element is $\text{Cov}(X_i, X_j)$. (Of course, this means that the diagonal holds the variances of the X_i .)

It is standard to let $\boldsymbol{\mu} = E(\mathbf{X})$ and $\boldsymbol{\Sigma} = V(\mathbf{X})$.

Some important properties: Let u and v denote two vectors of n scalars.

1. $E(\mathbf{u}^T \mathbf{X}) = \mathbf{u}^T E(\mathbf{X})$

$$2. V(\mathbf{u}^T \mathbf{X}) = \mathbf{u}^T \Sigma \mathbf{u}$$

$$3. \text{Cov}(\mathbf{u}^T \mathbf{X}, \mathbf{v}^T \mathbf{X}) = \mathbf{u}^T \Sigma \mathbf{v}$$

4. If $\mathbf{A}$ is p by n matrix, then $E(\mathbf{AX}) = \mathbf{A}\boldsymbol{\mu}$ and $V(\mathbf{AX}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T$

Exercise: Of course, the variance of a random variable cannot be negative.

What is an analogous constraint that is placed on Σ ?

[illegible]

Exercise: Suppose that X_1 , X_2 , and X_3 are random variables such that

$$E(X_1) = -1, \ E(X_2) = 0, \ E(X_3) = 2$$

and

$$V(X_1) = 1, V(X_2) = 3, V(X_3) = 1$$

and

$$\text{Cov}(X_1, X_2) = 0.5, \text{Cov}(X_1, X_3) = 0, \text{Cov}(X_2, X_3) = -0.5.$$

Let

$$Y_1 = X_1 - X_2$$

and

$$Y_2 = X_1 + 2X_2 + 9X_3.$$

What are

$E(Y_1)$, $E(Y_2)$, $V(Y_1)$, $V(Y_2)$, and $\text{Cov}(Y_1, Y_2)$?

[illegible]

Exercise: Let $\mathbf{X}$ be a random vector such that

$$E(\mathbf{X}) = \boldsymbol{\mu} = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

and

$$V(\mathbf{X}) = \boldsymbol{\Sigma} = \begin{bmatrix} 1 & 0.5 & 0 \\ 0.5 & 3 & -0.5 \\ 0 & -0.5 & 1 \end{bmatrix}.$$

Let

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 9 \end{bmatrix}.$$

What are the mean and variance of $\mathbf{Y} = \mathbf{A}\mathbf{X}$?

Joint Distributions

We can write $f_{\mathbf{X}}(\mathbf{x})$ for the **joint distribution** of $\mathbf{X}$, much as we did in the bivariate case.

For instance, if $\mathbf{X}$ is continuous,

$$P(X_1 \in (a_1, b_1), X_2 \in (a_2, b_2), \dots, X_n \in (a_n, b_n)) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} \cdots \int_{a_n}^{b_n} f_{\mathbf{X}}(\mathbf{x}) dx_n dx_{n-1} \cdots dx_1$$

We obtain multivariate marginal and conditional distributions as before:

Integrate out to get marginals. For example,

$$f_{(X_3, X_4, \dots, X_n)}(x_3, x_4, \dots, x_n) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{\mathbf{X}}(\mathbf{x}) dx_1 dx_2$$

For the conditionals **divide by density** for the random variables being conditioned on. For example,

$$f_{(X_1, X_2) | (X_3, X_4, \dots, X_n)}(x_1, x_2 | x_3, x_4, \dots, x_n) = \frac{f_{\mathbf{X}}(\mathbf{x})}{f_{(X_3, X_4, \dots, X_n)}(x_3, x_4, \dots, x_n)}$$

Exercise: Suppose that $X_1, X_2, \dots, X_n$ have joint pdf

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} \frac{2n!}{(2 + \sum_{i=1}^n x_i)^{n+1}}, & \text{if } x_i > 0, \ i = 1, 2, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

For this exercise, assume that $n = 5$.

1. What is the marginal distribution for (X_1, X_4) ?

2. What is the conditional distribution of (X_2, X_3, X_5) given (X_1, X_4) ?

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1 Independence

2 The random variables $X_1, X_2, \dots, X_n$ are independent if

$$3 \quad f_{\mathbf{X}}(\mathbf{x}) = \prod_{i=1}^n f_{X_i}(x_i) \quad \text{for all } x_1, x_2, \dots, x_n$$

4 When say that “ $X_1, X_2, \dots, X_n$ are i.i.d.” that means that they are **inde-**
 5 **pendent and identically distributed**. The term “identically distributed”
 6 means that the random variables have the exact same distribution, not sim-
 7 ply that they are “all normal” or “all exponential.”

1 Multivariate Normal

2 The vector $\mathbf{X}$ (of length n) has the **multivariate normal distribution** if it
 3 has joint pdf

$$4 \quad f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

5 It is assumed that Σ is positive definite.

6 This implies that $E(\mathbf{X}) = \boldsymbol{\mu}$ and $V(\mathbf{X}) = \Sigma$. All marginal and condi-
 7 tional distributions are multivariate normal. Further, any random vari-
 8 able/vector of the form $\mathbf{A}\mathbf{X}$ will also be multivariate normal.

9 If $\text{Cov}(X_i, X_j) = 0$, then X_i and X_j are independent, and if Σ is diagonal,
 10 then $X_1, X_2, \dots, X_n$ are independent.

Example: Portfolio Theory

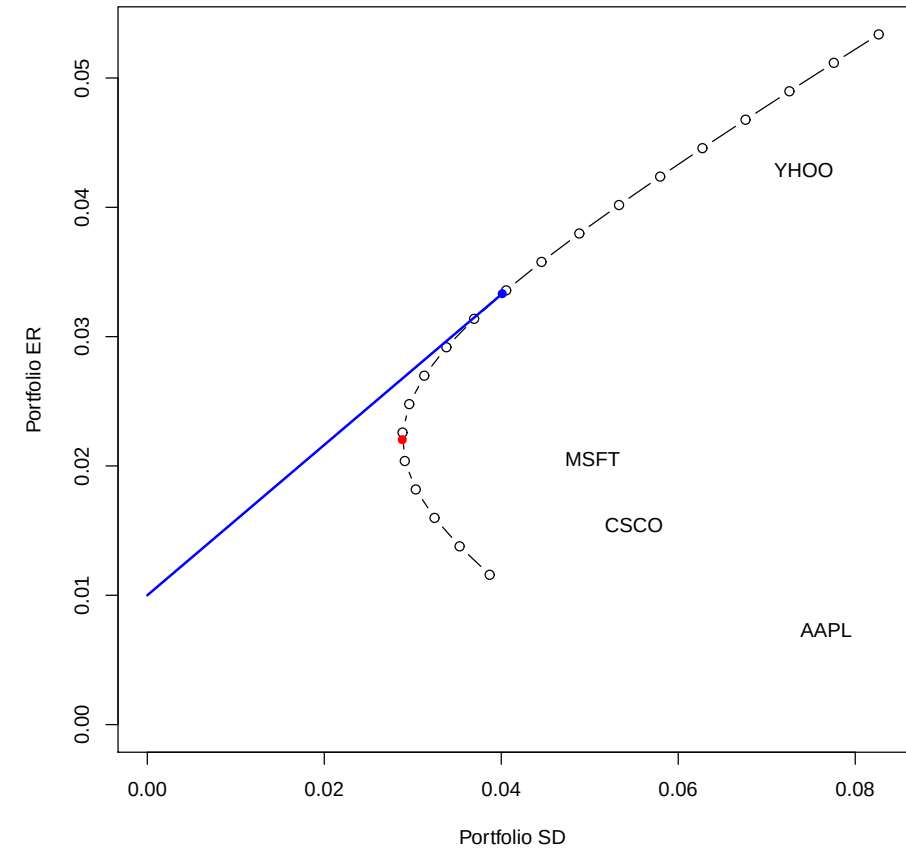
Let R_i denote the return on equity i for $i = 1, 2, \dots, n$, and let w_i denote the weight placed on equity i in the portfolio. Assume that $\sum_i w_i = 1$ and $w_i \geq 0$.

Let the overall return be $Y = \sum_i w_i R_i$.

Exercise: Describe the distribution of Y , and what properties an optimal portfolio would possess.

[illegible]

Efficient Frontier



Based on monthly returns from September 1, 2012 to September 1, 2014.